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Modeling Survival Data: Extending the Cox Model

With 80 Illustrations

Springer

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To Kathryn for her patience and support, to Victor for cracking the whip, and to Rex for nipping at our heels.

Preface

This is a book for statistical practitioners who analyse survival and event history data and would like to extend their statistical toolkit beyond the Kaplan-Meier estimator, log-rank test and Cox regression model to take advantage of recent developments in data analysis methods motivated by counting process and martingale theory. These methods extend the Cox model to multiple event data using both marginal and frailty approaches and provide more flexible ways of modeling predictors via regression or smoothing splines and via time-dependent predictors and strata. They provide residuals and diagnostic plots to assess goodness of fit of proposed models, identify influential and/or outlying data points and examine key assumptions, notably proportional hazards. These methods are now readily available in SAS and Splus.

In this book, we give a hands-on introduction to this methodology, drawing on concrete examples from our own biostatistical experience. In fact, we consider the examples to be the most important part, with the rest of the material helping to explain them. Although the notation and methods of counting processes and martingales are used, a prior knowledge of these topics is not assumed — early chapters give a not overly technical introduction to the relevant concepts.

SAS macros and S-Plus functions presented in the book, along with most of the data sets (all that are not proprietary) can be found on T. Therneau's web page at `www.mayo.edu/hsr/biostat.html`. It is our intention to also post any corrections or additions to the manuscript.

The authors would appreciate being informed of errors and may be contacted by electronic mail at

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Both authors would like to acknowledge partial support from DK34238-14, a long term NIH grant on the study of liver disease. The influence of this medical work on both our careers is obvious from the data examples in this volume.

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Patricia Grambsch
May 2000

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